

# Long-Distance Prediction of German Separable Verbs in Causal Language Models

Ahmed Amine Belloula · Abdullah Zafar

Universität Trier · M.Sc. Natural Language Processing  
12401633 – Trends in Natural Language Processing (2. Parallelgruppe) · Dr. phil Simon Munker

## 1. Motivation & Research Question

German separable verbs split their lexical components across a sentence:

aufstehen → *Ich stehe früh auf.*

The stem appears first, while the particle may occur many tokens later. This provides a natural test of **long-distance lexical integration**.

Prior work shows recurrent models degrade on long-distance subject–verb agreement as distance grows [4], and minimal-pair benchmarks such as BLiMP [9] formalize this evaluation style. Separable particles have mainly been studied for lemmatization and reconstruction [3, 1], not for long-distance resolution in language models.

**Research question:** Can causal language models preserve the correct stem–particle association as the particle is increasingly delayed?

**Hypothesis:** If lexical integration is sensitive to recency, particle prediction should deteriorate with distance.

## 2. Methodology

### Data

Naturally occurring sentences from **UD German GSD** [5] and **German HDT** [2].

- 17 verb stems
- 3 distance conditions
- 5 items per stem × condition
- 255 stimuli**

### Distance

| Condition | Intervening tokens |
|-----------|--------------------|
| Short     | ≤ 2                |
| Medium    | 3–7                |
| Long      | > 7                |

For each **compound:prt** instance we record the stem, gold particle, distance and sentence.

The sentence is truncated immediately before the particle. The gold particle is scored against alternative particles independently at-tested with the same stem.

**No synthetic stimuli or artificial distractors.**

## 3. Models & Evaluation

**Models.** Three GPT-2-style causal transformers, differing in scale and how German enters training [7]:

| Model        | Size | Type         | Tokenizer / Training                                            |
|--------------|------|--------------|-----------------------------------------------------------------|
| german-gpt2  | 124M | German       | Native German BPE, trained from scratch on German-only corpora  |
| gerpt2-large | 774M | German       | English GPT-2 tokenizer/weights, continued on German CC-100 [6] |
| mGPT         | 1.3B | Multilingual | Shared vocab across 61 languages [8]                            |

**german-gpt2** uses a native German tokenizer trained from scratch; **gerpt2-large** adapts the original *English* GPT-2 (English tokenizer/weights) to German; **mGPT** shares its vocabulary across 61 languages. This lets us test whether *how* German was learned — not just model size — predicts performance.

For each candidate particle  $p$ , teacher-forced scoring computes:

$$\log P(p \mid c) = \sum_t \log P(p_t \mid c, p_{<t})$$

**Metrics.** Let  $c_1 > c_2 > \dots > c_k$  be candidate log-probabilities sorted in descending order, with the correct particle  $p^*$  at rank  $r$ .

- Top-1 accuracy:** fraction of items with  $r = 1$ , i.e.  
 $\log P(p^* \mid c) = \max_i \log P(p_i \mid c)$ .
- MRR:**  $\frac{1}{N} \sum_{n=1}^N \frac{1}{r_n}$  — more informative than accuracy with up to 44 candidates, since it rewards a correct particle ranked 2nd differently from one ranked 40th.
- Margin:**  $\log P(p^* \mid c) - \max_{i \neq *} \log P(p_i \mid c)$ , the log-probability gap between the correct particle and its strongest distractor; positive margin means the model prefers the correct particle.

## 4. Results: Distance

### Accuracy

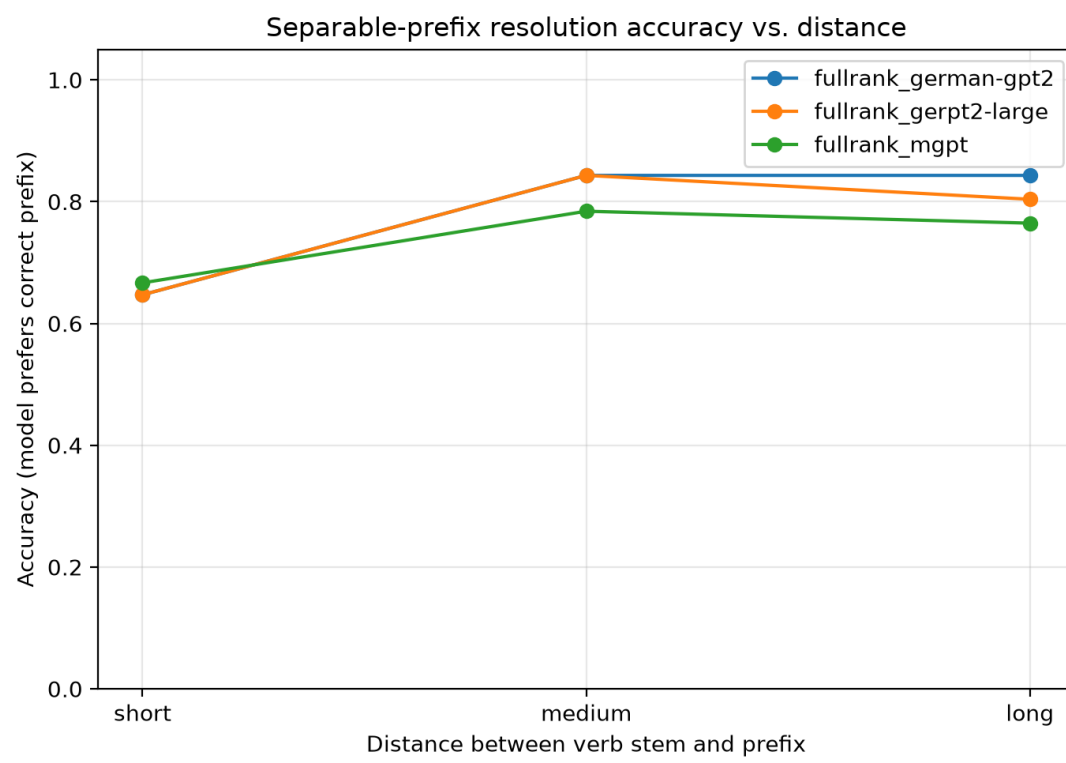


Figure 1. Top-1 particle prediction accuracy.

### Ranking performance

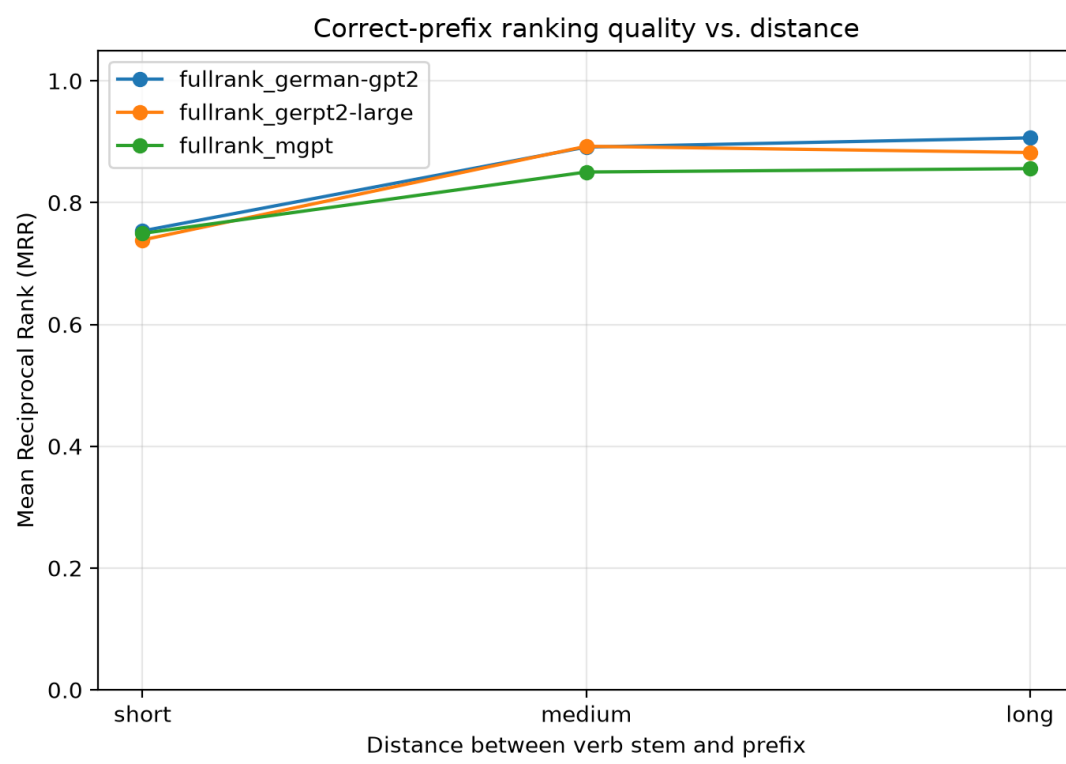


Figure 2. Mean reciprocal rank.

### Log-probability margin

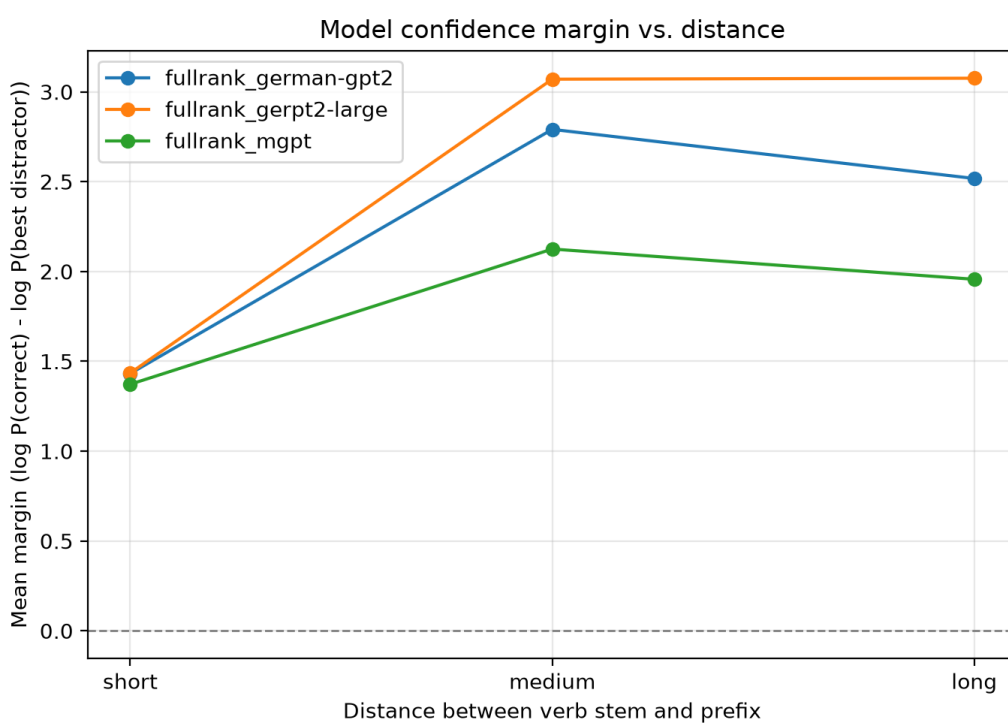


Figure 3. Correct-vs-best-distractor log-probability margin.

Positive margin means that the model prefers the correct particle over its strongest corpus-attested alternative.

### Main observation

Across the three models, performance rises from short to medium distance and then largely plateaus or slightly decreases at long distance.

## 5. Interpretation

The expected monotonic distance penalty is not observed. The correct particle remains highly predictable even when the stem–particle dependency becomes longer — directly contrasting with the recency-driven degradation reported for LSTMs on subject–verb agreement [4].

The dip at short distance is more plausibly explained by **insufficient preceding context** to disambiguate the verb's sense, rather than by distance itself — short-distance items simply offer the model less material to work with before the particle appears. This is consistent with the jump in accuracy from short to medium distance, though our within-bucket context-length check (§7) finds no strong effect once distance is already fixed, suggesting the relevant factor is the *presence* of disambiguating context rather than its exact length.

This suggests the tested models maintain useful information about the delayed particle across the ranges tested here, while other lexical properties may contribute more to prediction difficulty than raw distance — a possibility examined directly in the confound checks below (§7).

## 6. Lexical Preference

### Lexical variation

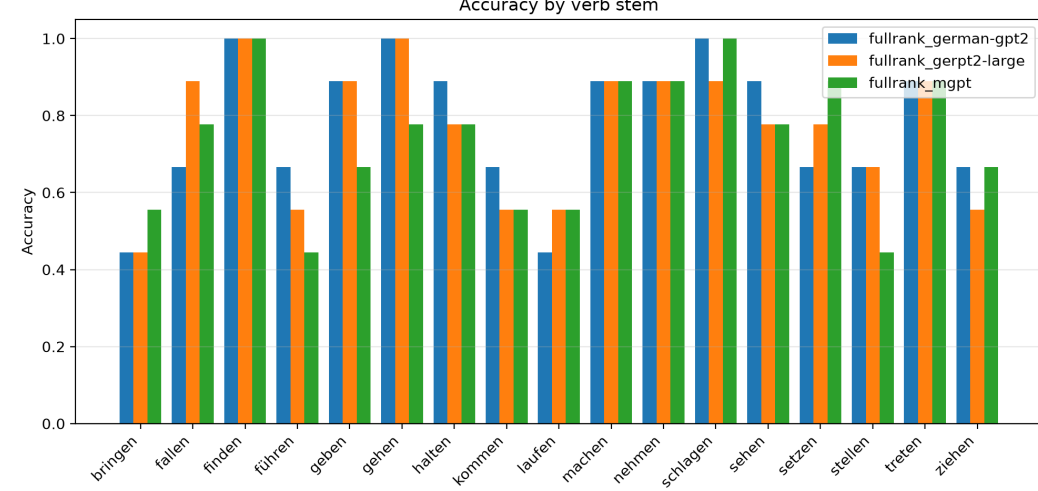


Figure 4. Accuracy varies across verb stems.

*finden* and *gehen* are near ceiling, while *bringen* and *laufen* show lower accuracy.

## 7. Confound Checks

### Candidate-set size

| Model        | Pearson r |
|--------------|-----------|
| german-gpt2  | 0.004     |
| gerpt2-large | −0.058    |
| mGPT         | −0.320    |

Candidate-set size shows little linear relationship with accuracy for the two German-specific models; mGPT shows a weak negative relationship, suggesting mild sensitivity to larger competitor pools.

### Context length

Pearson correlation between context length and accuracy, within distance buckets ( $n=51/\text{bucket}$ ):

$$|r| < 0.23$$

indicating weak and inconsistent relationships between exact context length and accuracy.

## 8. Conclusion

Contrary to the recency-driven degradation reported for LSTMs on subject–verb agreement [4], increasing stem–particle distance does **not** produce a simple monotonic decline in particle prediction.

**Contribution.** A fully automatic, corpus-grounded pipeline — no model training, no manual dataset construction — combined with full-ranking evaluation against every attested competitor, providing the first test of this meaning-critical long-distance dependency in transformer LMs.

The results instead point toward a combination of **distance and lexical-specific difficulty**.

**Limitations:** 17 stems / 255 stimuli; models tested range from 124M–1.3B parameters, evaluated locally under free-compute constraints — no frontier-scale models were tested; distractors differ in semantic similarity to the correct answer, not explicitly controlled; linear distance captures only one aspect of dependency difficulty.

## References

- [1] Dolores Batinić and Thomas Schmidt. Reconstruction of separable particle verbs in a corpus of spoken german. In Georg Rehm and Thierry Declerck, editors, *Language Technologies for the Challenges of the Digital Age*, pages 3–10, Cham, 2018. Springer International Publishing.
- [2] Emanuel Borges Völker, Maximilian Wendt, Felix Hennig, and Arne Köhn. HDT-UD: A very large Universal Dependencies treebank for German. In Alexandre Rademaker and Francis Tyers, editors, *Proceedings of the Third Workshop on Universal Dependencies (UDW, SyntaxFest 2019)*, pages 46–57, Paris, France, August 2019. Association for Computational Linguistics.
- [3] Maximilian Köper, Sabine Schulte im Walde, Max Kisselew, and Sebastian Padó. Improving zero-shot-learning for German particle verbs by using training-space restrictions and local scaling. In Claire Gardent, Raffaella Bernardi, and Ivan Titov, editors, *Proceedings of the Fifth Joint Conference on Lexical and Computational Semantics*, pages 91–96, Berlin, Germany, August 2016. Association for Computational Linguistics.
- [4] Tal Linzen, Emmanuel Dupoux, and Yoav Goldberg. Assessing the ability of LSTMs to learn syntax-sensitive dependencies. *Transactions of the Association for Computational Linguistics*, 4:521–535, 2016.
- [5] Ryan McDonald, Joakim Nivre, Yvonne Quirmbach-Brundage, Yoav Goldberg, Dipanjan Das, Kuzman Ganchev, Keith Hall, Slav Petrov, Hao Zhang, Oscar Täckström, Claudia Bedini, Núria Bertomeu Castelló, and Jungmee Lee. Universal Dependency annotation for multilingual parsing. In Hinrich Schuetze, Pascale Fung, and Massimo Poesio, editors, *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 92–97, Sofia, Bulgaria, August 2013. Association for Computational Linguistics.
- [6] Benjamin Minixhofer. GerPT2: German large and small versions of GPT2, 12 2020.
- [7] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [8] Oleh Shliachko, Alena Fenogenova, Maria Tikhonova, Anastasia Kozlova, Vladislav Mikhailov, and Tatiana Shavrina. mGPT: Few-shot learners go multilingual. *Transactions of the Association for Computational Linguistics*, 12:58–79, 2024.
- [9] Alex Warstadt, Alicia Parrish, Haokun Liu, Anhad Mohanoney, Wei Peng, Sheng-Fu Wang, and Samuel R. Bowman. BLiMP: The benchmark of linguistic minimal pairs for English. *Transactions of the Association for Computational Linguistics*, 8:377–392, 2020.